

2026 Game Day Analytics Challenge

White Paper



Team 9 - $e^{i\pi} + 1 = 0$

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1. Introduction

On February 8, 2026, Super Bowl LX was held at Levi's Stadium in Santa Clara, California, drawing millions of viewers from across the United States and around the world. As one of the most widely watched live sporting events globally, the Super Bowl serves not only as a championship game but also as a premier platform for large-scale advertising and brand exposure. During this year's broadcast, 66 commercials were aired, including 18 from first-time advertisers. Total advertising expenditures were estimated at approximately \$550 million, with an average cost of about \$8 million for a 30-second spot.

This white paper analyzes social media activity related to Super Bowl LX advertisements using a dataset of tweets collected during the event. By examining engagement metrics, temporal patterns, viral content characteristics, and brand mentions, the study aims to provide insight into how audiences responded to advertisements in real time. Understanding these dynamics can help advertisers, marketers, and researchers evaluate campaign effectiveness and develop data-driven strategies for future large-scale events.

2. Methodology & Technical Approach

This section integrates the analytical procedures and key findings derived from the Super Bowl LX tweet dataset. All analyses were conducted using Python in a Google Colab environment on a cleaned dataset of social media posts related to advertisements aired during the event.

2.1 Tech Stack & Tools

Our team used a cloud-based Python environment to ensure reproducibility and collaboration.

- **Environment:** Google Colab integrated with Google Drive for cloud storage and data ingestion.
- **Core Languages:** Python
- **Data Manipulation:** Pandas for dataframe management, cleaning, and aggregation, NumPy for numerical operations.
- **Text Processing:** Re (Regular Expressions) for pattern matching, Collections (Counter) for frequency analysis of hashtags.

2.2 Data Preprocessing & Cleaning

We implemented a strict cleaning protocol to ensure statistical validity before analysis began:

- **Duplicate Handling:** We found massive duplication of specific tweet IDs in the raw dataset (final-gameday[main].csv). These duplicates would have artificially inflated engagement metrics. Therefore, we implemented a strict drop_duplicates logic based on the unique identifier, keeping only the first occurrence. This process eliminated statistical noise and prevented double-counting of likes and impressions.
- **Missing Value (NA) Imputation:** Rows missing critical data that include id and text were dropped immediately as they hold no analytical value. Nulls in metric columns like public_metrics were filled with 0. We assumed absence of

data equates to zero engagement rather than eliminating the tweet. Nulls in metadata like location or geo were filled with unknown. We retained these rows to keep the sentiment data contained in the text, avoiding the loss of valid user opinions due to missing location tags.

- **Data Typing:** We converted tweet_id, author_id, and conversation_id to strings to prevent numerical corruption and to disable mathematical operations on identifiers. We also converted created_at to datetime objects to enable time-series analysis.

2.3 Data Transformation & Entity Resolution

To link tweets to Super Bowl commercials accurately, we developed a hybrid brand detection algorithm rather than a simple keyword match.

- **Hybrid Mapping Strategy**
 - Hashtags: We mapped tweets using a verified list of official campaign hashtags sourced from ad_keywords.csv.
 - Text Search: If no hashtag was found, we performed a strict text search for the brand name, considering variations (e.g., “Lay’s” vs “Lays”).
- **Semantic Filtering**
 - The Brand “Spectrum” Problem: The word "Spectrum" is too generic. We implemented a context filter that only assigned the brand “Spectrum” if the text also contained service-related keywords or competitor mentions.
 - The Brand “MAHA” Problem: To detect the official ad, we required the presence of the #maha hashtag or context words related to the commercials to distinguish the movement from phonetically similar words.
- **Refinement of the Unmapped Hashtags:** We analyzed the top 100 most frequent unmapped hashtags to find missing signals. We found missing hashtags and manually mapped to the corresponding brand.

- **Merge Brands:** We identified that the brand “PepsiCo” and “Pepsi Zero Sugar” referred to the same commercial. We merged these fragmented brand identifiers into a single master entity: Pepsi. This ensured the brand’s total impact was not fragmented across multiple labels.

3. Visualization & Insights

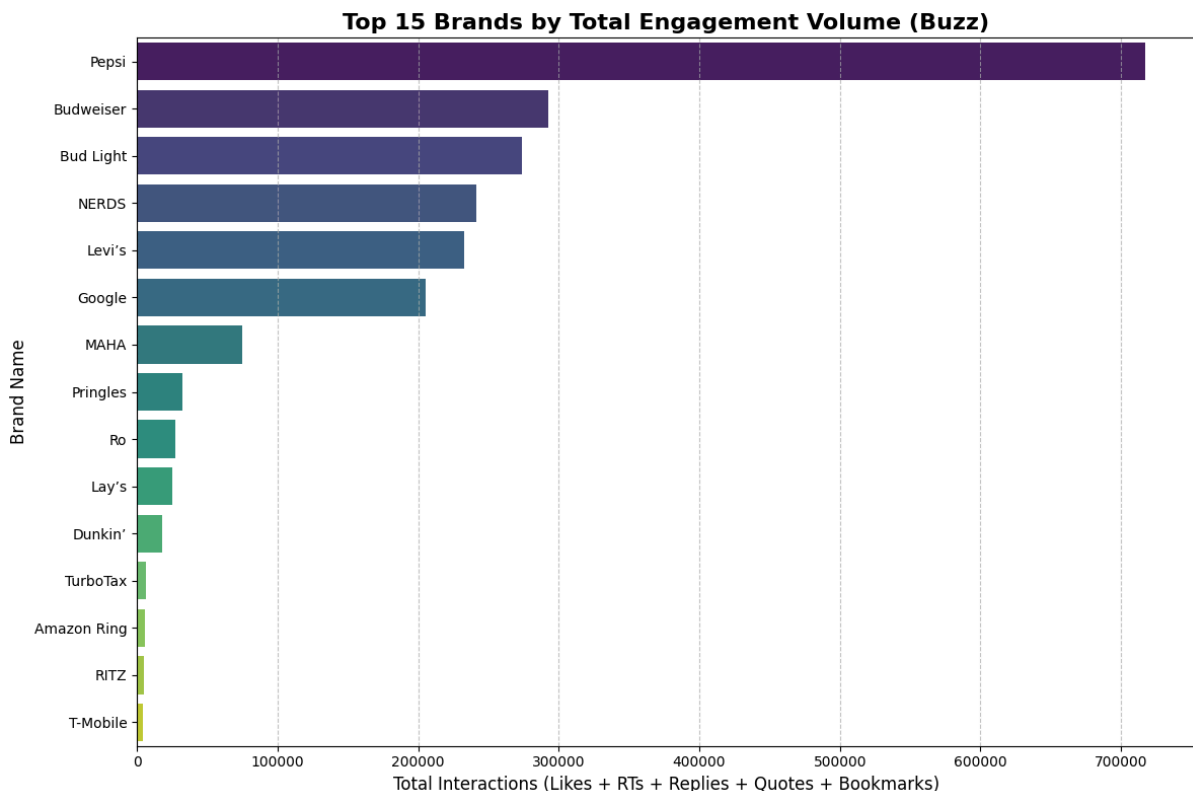
3.1 Engagement & Volume

Engagement Analysis

The success of Super Bowl advertisements is not simply determined by how many people have watched them. The traditional indicator, impressions, shows how widely an advertisement was exposed, but this represents only passive exposure. Today, Superbowl advertisements have become a social event that is consumed and reinterpreted online in real time beyond watching television.

In this analysis, engagement is defined as the sum of likes, retweets, replies, quote tweets, and bookmarks. This reflects not only simple exposure but audience interaction and indicates how strongly an advertisement elicits viewers' emotions and actions. In other words, engagement measures whether viewers reacted to the advertisement, rather than whether they merely saw it.

3.1.1 Total engagement volume



The distribution of total engagement volume shows that **Pepsi** generated the **highest level of interactions**, followed by Budweiser. This pattern suggests that

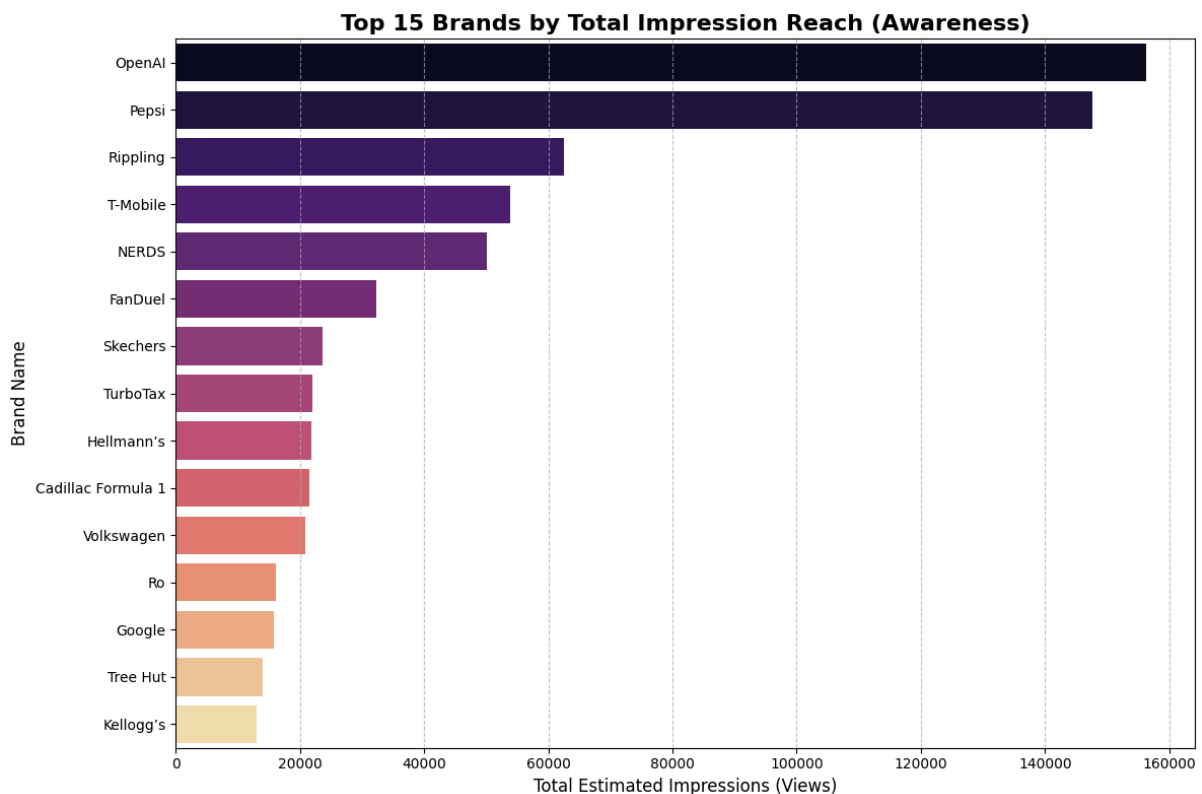


Super Bowl advertising operates as a competition rather than equal exposure across advertisers.

Although Pepsi and Budweiser both recorded high interaction columns, the results should not be interpreted as a direct measure of advertisement quality. Pepsi encouraged participation through **interactive social media activities**, while Budweiser's engagement was primarily driven by **audience reaction to the advertisement content**.

This comparison suggests that engagement reflects multiple factors, including both **creative impact** and **platform activation**.

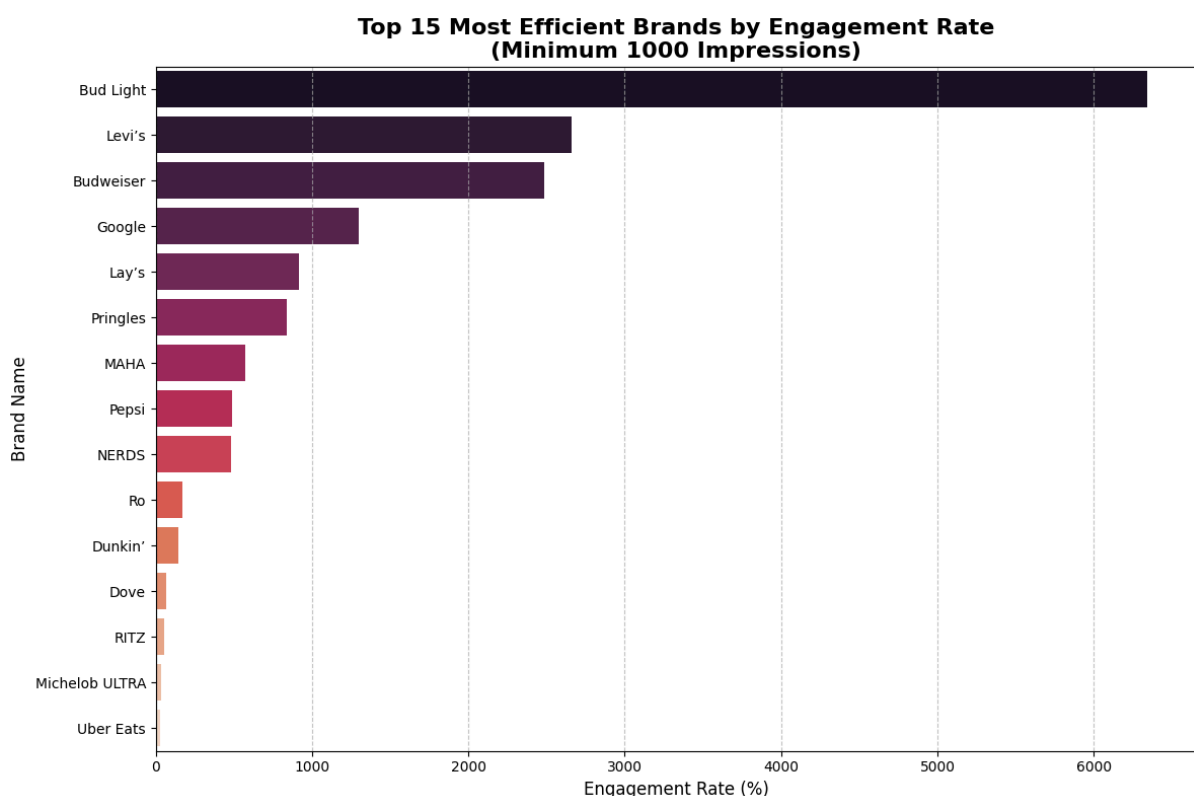
3.1.2. Impression Reach



To assess visibility, we analyzed impression reach, which measures the number of users who watched brand-related content. The results show that perception and interaction did not match perfectly. OpenAI had the highest impressive reach, followed by Pepsi.

Although the difference was not significant, the ranking differs from the engagement results, where Pepsi led in total interactions. This suggests that exposure and interaction do not necessarily move in the same direction. Even if visibility levels are similar, audience reactions may vary depending on campaign structure and content appeal. Thus, reach reflects attention potential, while engagement reflects response intensity.

3.1.3 Engagement rate



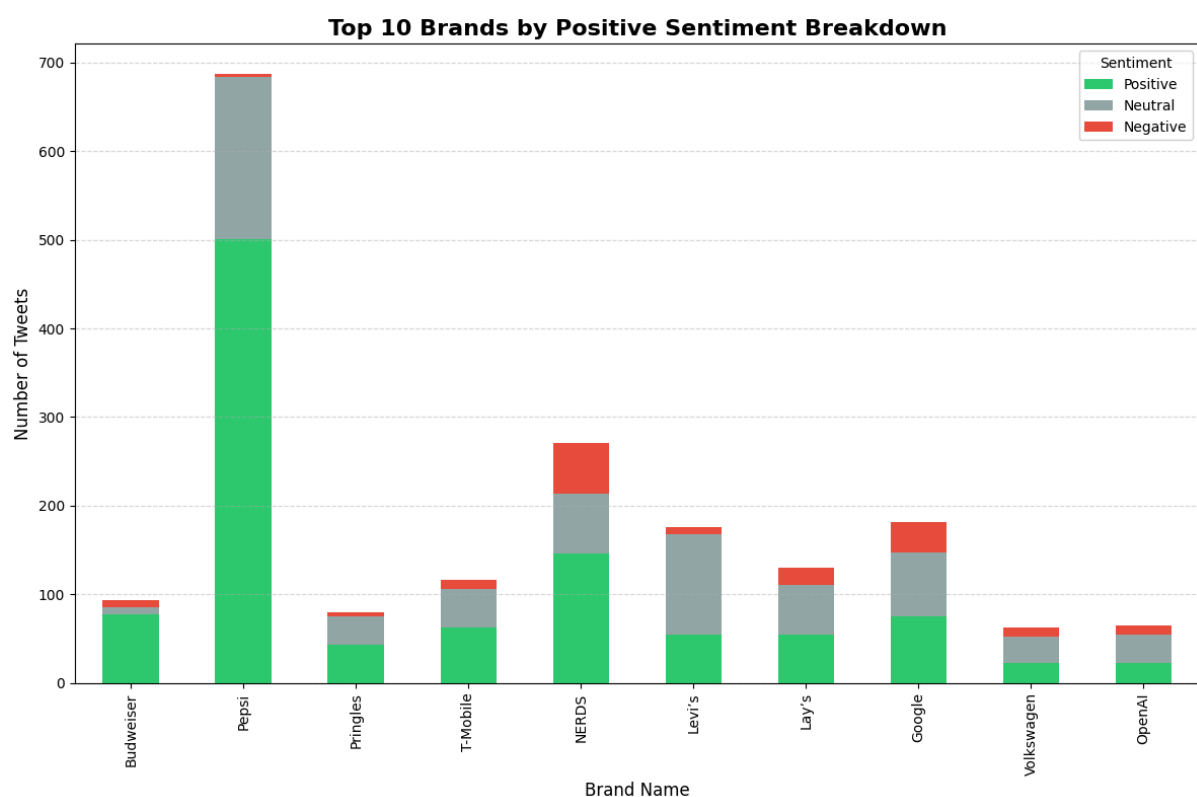
We calculated the Weighted Engagement Rate (Total Interactions ÷ Impression Reach) to generate the most intense response per viewer. This metric shows the “Viral Density” of a commercial. **Bud Light** emerged as the overwhelming leader in efficiency with a staggering engagement rate of 6,343%. A rate exceeding 100% indicates that the viewer interacted multiple times or that the content sparked secondary conversations that outpaced its initial reach. While Bud Light had a

relatively low reach (4,313), the intensity of the response was unmatched. This shows a highly concentrated activation where a specific audience segment became vocal advocates or critics of the brand. **Pepsi** recorded a 486% engagement rate despite its massive volume. This is still exceptionally high but demonstrates the difficulty of maintaining such intense interactivity at a mass scale.

3.2 Sentiment & Emotion

We used the NLTK VADER tool to label tweets as positive, negative, or neutral. Then, we calculated a “Net Sentiment Score” to see which brands were most loved. To go deeper, we used NRCLEx to find specific emotions like joy, trust, and added our own custom search to measure ‘Nostalgia’. Finally we combined these into an “Emotional Resonance Score” and created a heatmap to visualize exactly how fans felt about each brand.

3.2.1 Net Sentiment Score: The “Brand Love” Metric

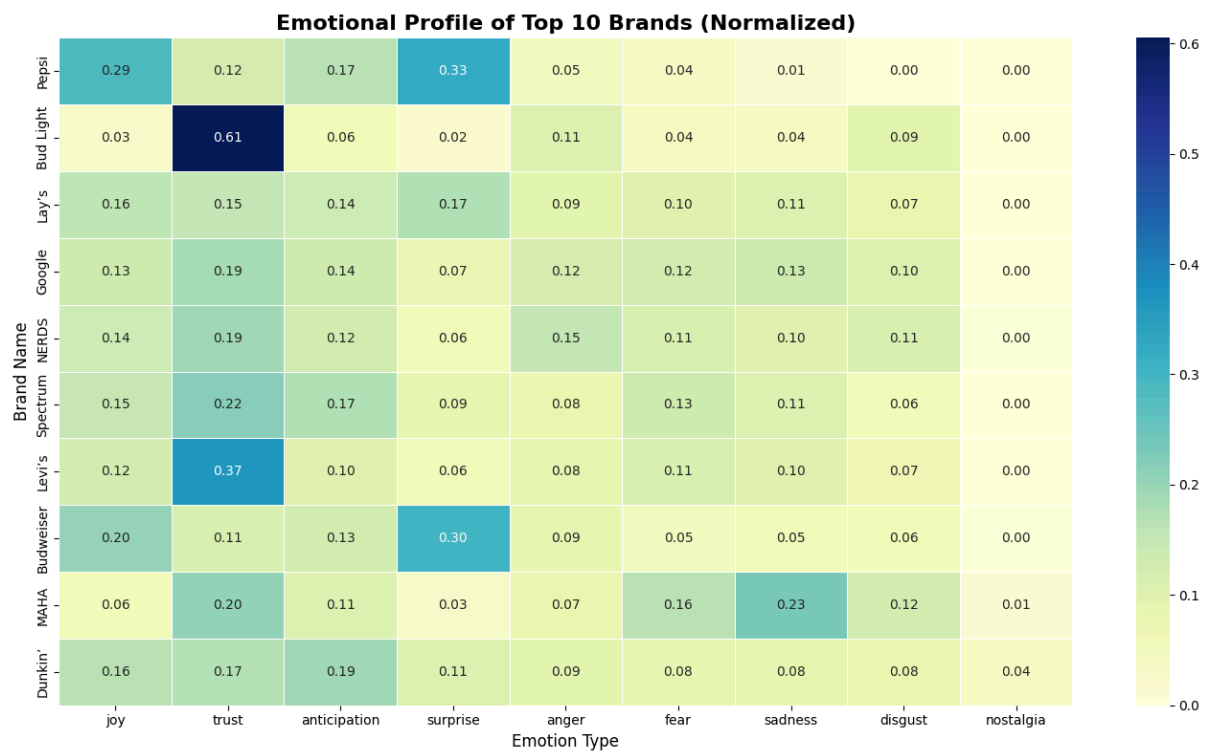


We calculated the Net Sentiment Score (positive% - negative%) to identify which brands were genuinely loved versus those that simply generated noise.

Pepsi achieved **high-volume consistency** with a 72.5% net sentiment score. This is impressive because they had a massive volume of 687 tweets. 72.9% of these tweets were positive, and less than 1% were negative. Pepsi showed that a brand can reach a huge audience without making people angry. **Budweiser** achieved **peak positivity**. Budweiser had fewer tweets (93) than Pepsi. However, they achieved the highest net sentiment score with 74.2%. Their commercial generated a huge 82.8% positive response. This shows their ad was safe and appealed to almost everyone.

Nerds had a high total volume (271 tweets). However, they also had a high negative ratio (21%). This brought their net sentiment score down to 32.8%. The data shows their commercial grabbed attention, but it split the audience. Some people liked it, but it alienated others.

3.2.2 Emotional Resonance: Beyond Good vs. Bad

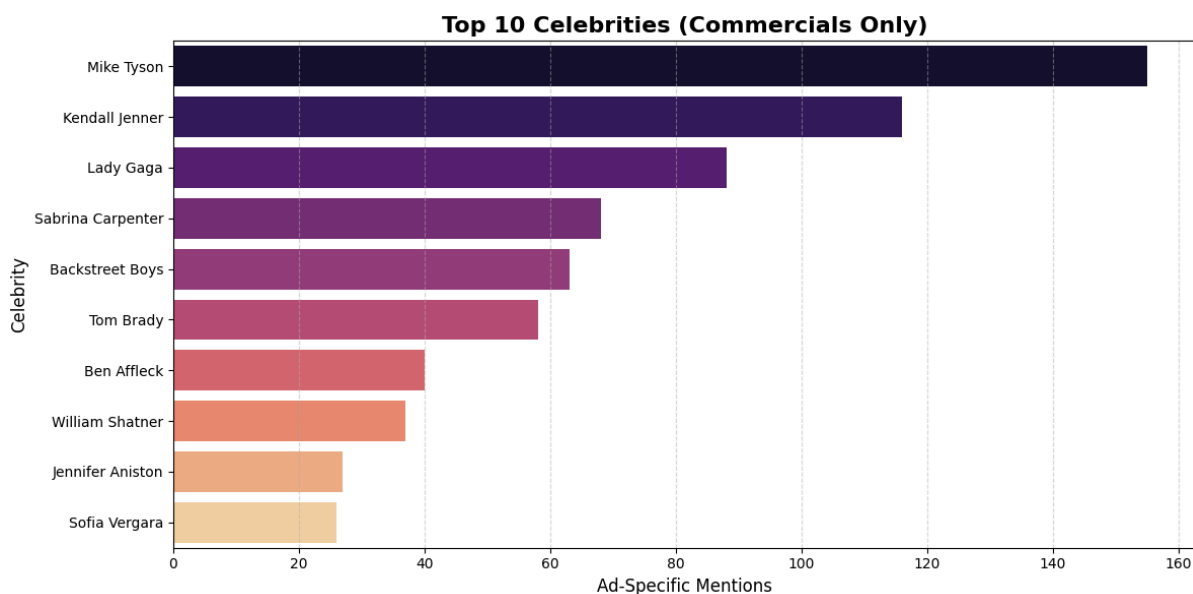


We mapped specific emotions to understand exactly how the commercials made viewers feel. This revealed different emotional profiles for each brand. The heatmap shows that **Pepsi** showed comparatively higher levels of **Joy** and **Surprise** categories. This makes sense because their ad was high-energy and entertaining. It also matches their status as the leader in total interactions (717k). Fewer people talked about **Bud Light** than Pepsi, but it achieved the highest **Trust** score. This matches their #1 ranking in Weighted Engagement Rate. The insight is fewer people posted about Bud Light, but those who did felt a strong, positive connection to the message. In contextual negativity, the “Make America Healthy Again” (**MAHA**) campaign triggered high levels of **Sadness** and **Fear**. This does not mean the audience hated the commercial, but it reflects the ad’s serious story. The commercial talked about health and poison in food, which evoked more serious or concern-related reactions among users.

We created a custom “Nostalgia” score. We wanted to see if retro or old-school themes were the main reason for success in Super Bowl 2026. However, nostalgia had almost no impact on the results this year. The scores were very low across the board. The brand with the highest Nostalgia score was **RITZ**(11). **Dunkin** followed with a score of 6. High nostalgia scores did not lead to brand love, high volume, or high efficiency. This shows that relying on the “Nostalgia” theme was not a winning strategy for mass appeal or engagement in this year’s Super Bowl.

3.2.3 Celebrity Buzz

To accurately measure celebrity impact, we applied a “Strict Mode” filter. Super Bowl stars often appear in Halftime Show or attend the game as fans. To avoid counting these non-commercial mentions, we rejected any tweet containing words like halftime, or show. We only counted tweets that explicitly mentioned “ad,” “commercial,” or the specific brand.



Mike Tyson was the #1 most mentioned celebrity in the commercials, with 155 verified mentions for the **MAHA** campaign. Kendall Jenner ranked #2 with 116 mentions, representing **Fanatics Sports**. Lady Gaga ranked #3 with 88 mentions for her role with **Rocket Mortgage & Redfin**. Interestingly, the brands with the highest net sentiment, which are Budweiser and Pepsi, did not rely on those top-tier celebrity

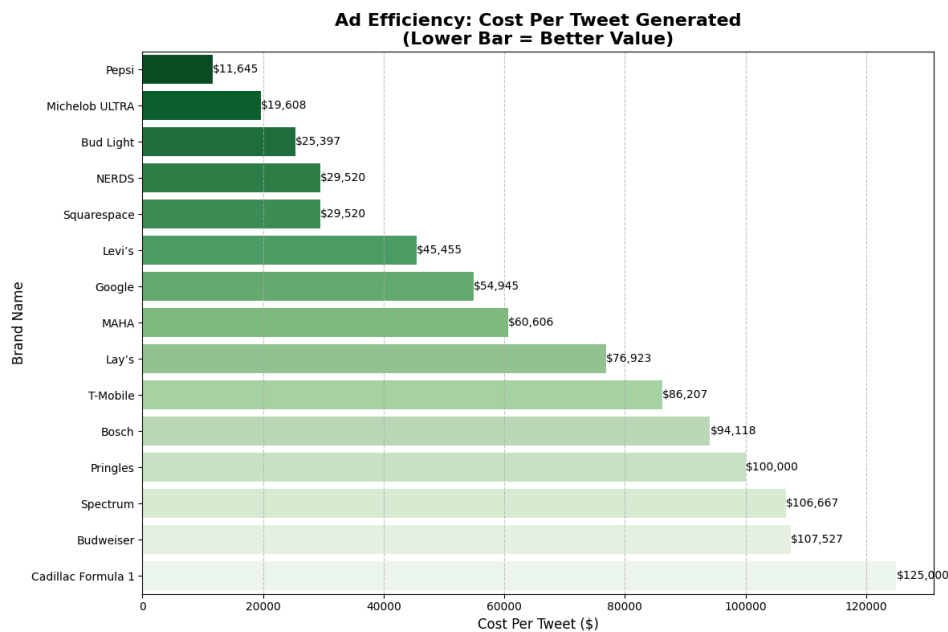
cameos to drive their success. Neither Budweiser nor Pepsi relied on the specific “Top 10 Most Mentioned Celebrities” to drive their high sentiment score.

While Mike Tyson successfully drove engagement volume, his presence did not necessarily drive positivity. The MAHA campaign, despite being efficient, triggered specific negative emotions (3.2.2). This demonstrates that while celebrities generated personal buzz, they failed to propel their brands into the top engagement tiers, showing that creative strategy is more effective than star power.

3.3 Business Efficiency

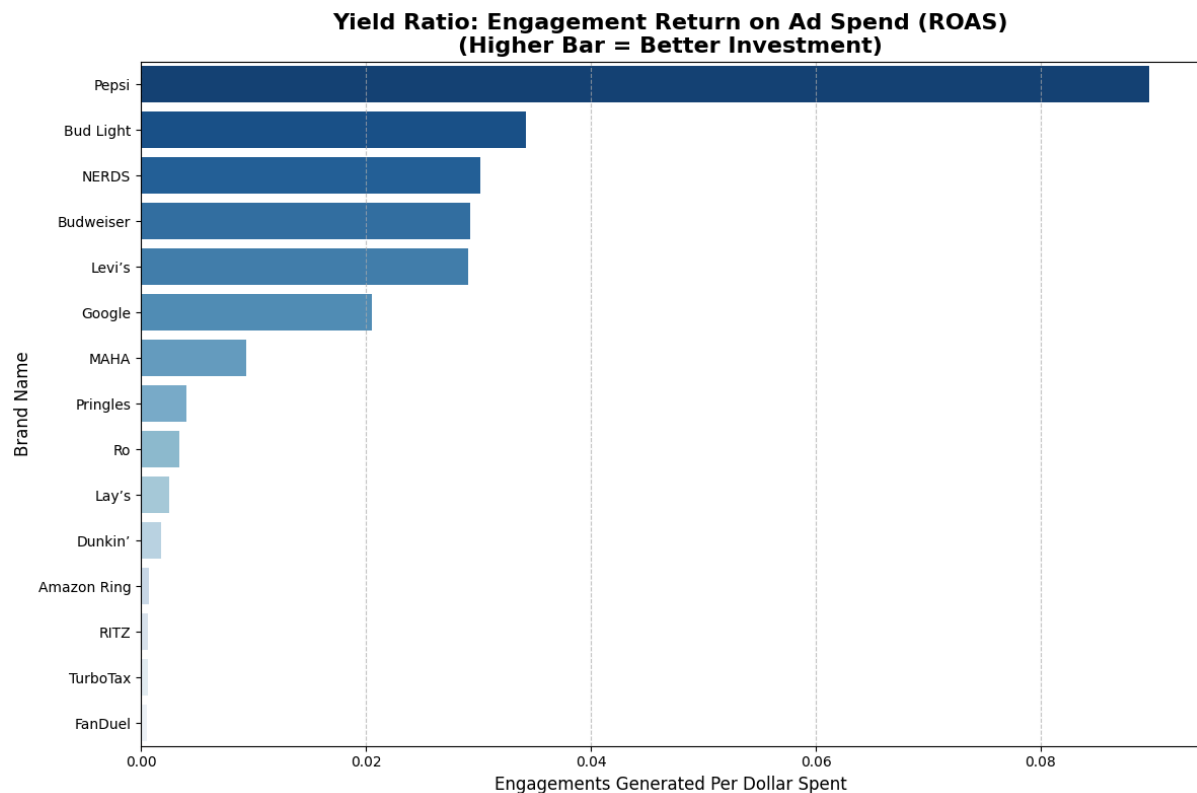
We used 3 different performance metrics to evaluate business efficiency. We used Cost Efficiency (Total Ad Spend ÷ Total Tweet Count), Yield Ratio (Total Engagements ÷ Total Ad Spend), and Stock Market Correlation (Compared stock market change between Feb 6 and Feb 9).

3.3.1 Cost Efficiency



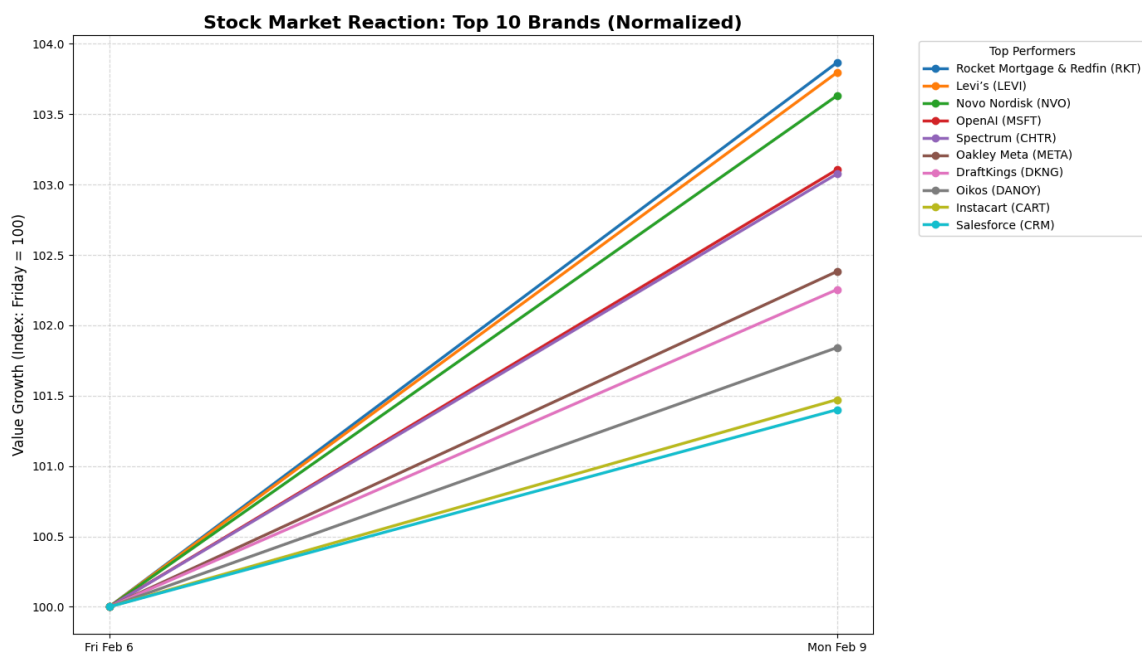
Pepsi emerged as the dominant performer in social media cost-effectiveness. In terms of **Cost Efficiency**, Pepsi recorded the most favorable figure at **\$11,645 per tweet**. This represents a significant margin, costing approximately \$8,000 less per tweet than the runner-up, Michelob ULTRA.

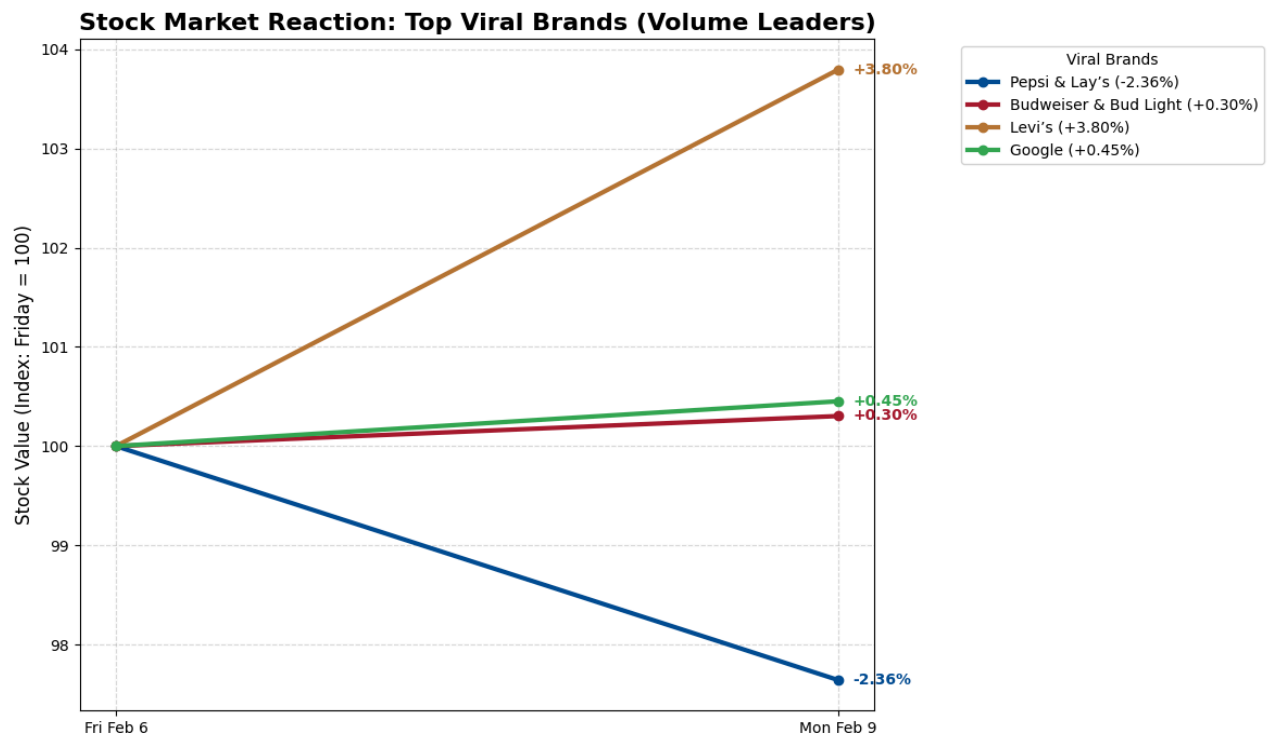
3.3.2 Yield Ratio



Pepsi also led the Yield Ratio rankings with a score of **0.0897**, demonstrating superior engagement ROI. This ratio was approximately **2.62 times higher** than that of the second-ranked brand, **Bud Light**.

3.3.3 Stock Market Correlation





Stock price data was sourced using the yfinance API, which retrieves historical market data from Yahoo Finance. We analyzed the immediate market reaction by comparing the closing stock prices from Friday, February 6, 2026 (the last trading day before the Super Bowl) to Monday, February 9, 2026 (the first trading day after). The percentage change was calculated for each ticker to rank performance. The starting point is set to 100 for all brands because the data has been normalized (or indexed) to allow for a fair comparison. For distinct brands operating under a single parent company, such as Pepsi and Lay's (PepsiCo) or Bud Light and Budweiser (Anheuser-Busch InBev), we tracked the performance of the parent entity's stock ticker. This approach attributes the parent company's aggregate market fluctuation to each of its subsidiary brands. We excluded private companies and non-profit organizations from this analysis as they do not have publicly traded stock tickers available for tracking.

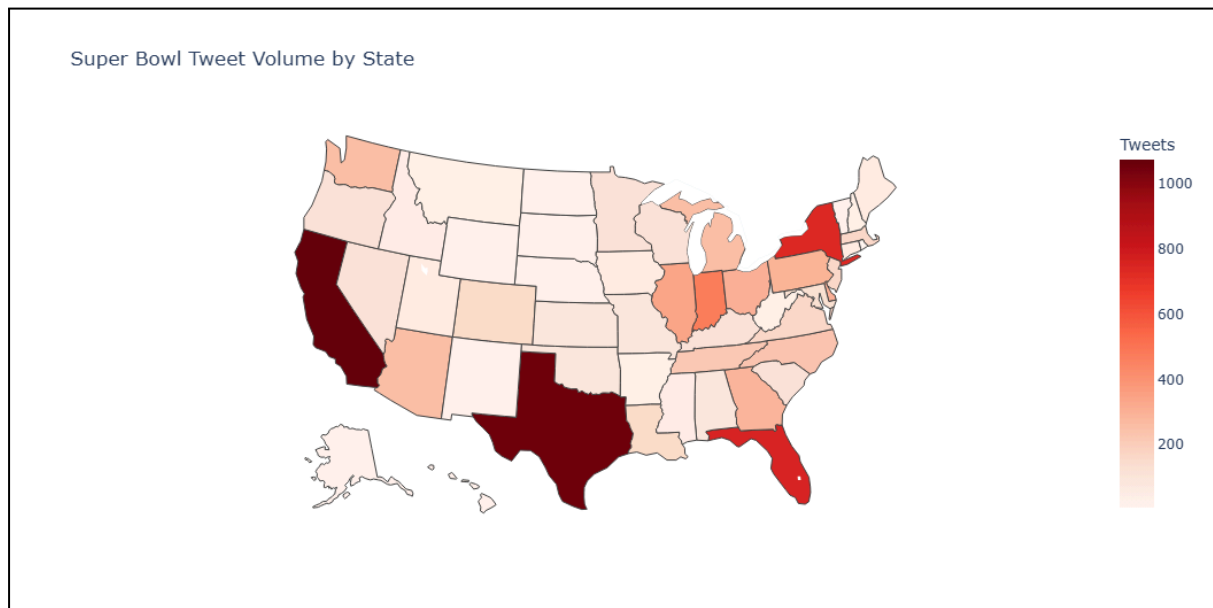
Despite Pepsi's overwhelming success in social metrics, this dominance did not translate into immediate stock market gains. Surprisingly, **Pepsi's stock price** experienced a **2.36% decline** on February 9 compared to February 6. Conversely, the highest stock appreciation was observed in **Rocket Mortgage & Redfin**, which posted an increase of approximately **3.87%**.

The first figure displays the top 10 companies with the highest stock price increases from February 6 to February 9, while the second figure illustrates the market performance of the top 10 'Viral Brands' identified by tweet volume. A comparative analysis of these two datasets reveals a distinct disconnect: high social media visibility during the Super Bowl did not correlate with immediate stock market gains. This suggests that while viral campaigns successfully drive public conversation, they do not guarantee a short-term impact on shareholder value.

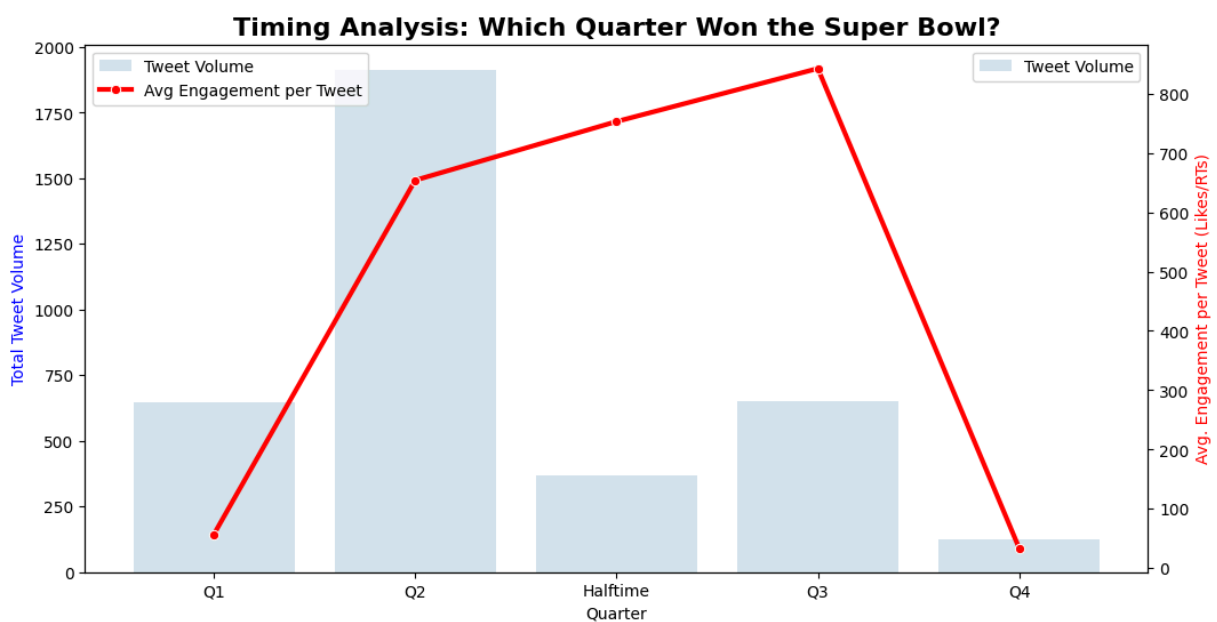
3.3.4 Summary

The disparity between social performance and stock trends suggests that high consumer influence, such as tweet volume and yield ratios does not guarantee a direct or immediate positive impact on stock prices. It is plausible that the three day observation wouldn't be sufficient for the generated public sentiment to fully materialize within the financial markets, or that social buzz operates independently of short-term investor behavior.

3.4 Contextual & Advanced Analytics



The geographic and timing analyses together show that Super Bowl advertising impact varied significantly across both location and broadcast period. Tweet activity was concentrated in a small number of states—particularly large, highly populated regions such as **California, Texas, Florida, and New York**—indicating that national discussion was driven primarily by major metropolitan areas rather than evenly distributed across the country.



At the same time, audience response fluctuated throughout the game. The **second** quarter generated the **highest tweet volume**, reflecting peak viewership and early

excitement, while the **third** quarter produced the **highest average engagement per tweet**, suggesting more focused interaction among viewers who remained actively engaged. In contrast, the **fourth** quarter showed both **low volume and low engagement**, indicating declining attention as the game concluded.

Together, these findings suggest that advertising effectiveness depends strongly on contextual factors: where audiences are located and when ads are aired. Early placements maximize reach and discussion, while later placements may yield stronger engagement among a smaller but more attentive audience.

4. Discussion

The analysis demonstrates that successful Super Bowl advertising cannot be explained by a single metric. Engagement, impression reach, sentiment, and efficiency represent different stages of audience response and do not necessarily align. High visibility increased awareness, but it did not consistently produce interactions, and high interaction did not always correspond to positive perception. These results show that advertising impact involves not only attracting attention but shaping how audiences react and feel about the message.

Performance was influenced by strategy and context than by scale. Higher spending and social media visibility did not guarantee stronger results or immediate financial impact. Instead, campaigns that combined emotional appeal, clear storytelling, and effective timing performed better across several metrics.

Moderately priced campaigns were able to outperform higher-budget advertisements when their content resonated with audiences, demonstrating the value of social media analytics in evaluating campaign effectiveness after broadcast.

5. Conclusion

This study provides a comprehensive evaluation of Super Bowl LX advertising performance using large-scale social media data. By integrating engagement metrics, sentiment analysis, emotional profiling, celebrity influence, cost efficiency, market response, geographic distribution, and timing analysis, the research identifies key drivers of advertising success during major live events.

Key conclusions include:

- Engagement is concentrated among a small number of standout campaigns
- Emotional resonance strongly influences audience response
- High visibility does not guarantee interaction
- Efficient campaigns can outperform high-budget advertisements
- Advertising timing affects both reach and engagement
- Celebrity endorsements can significantly amplify discussion

These findings highlight the importance of data-driven strategies for optimizing advertising effectiveness in high-visibility environments.

6. Limitations of Analysis

- Since the initial dataset was unstructured, we performed pre-processing, such as eliminating duplicates and filtering out bot-driven content. The results changed drastically after these cleaning steps, which clearly indicates that the quality of data pre-processing is a major determinant in the accuracy of our analysis.
- Brand detection relied on a hybrid keyword/hashtag mapping strategy. This approach captures explicit mentions but misses implicit references. Consequently, brands with complex spelling or generic names may have slightly underreported volume compared to brands with unique identifiers.
- Stock prices are influenced by macroeconomic factors, earning reports, and broader market volatility. Attributing price movements solely to Super Bowl advertising without a control variable or a longer longitudinal study prevents us from establishing a definitive causal link between social buzz and shareholder value.

7. Research References

- Super Bowl LX Tweet Dataset
- VADER Sentiment Analysis Model
- Yahoo Finance Market Data
- <https://sports.yahoo.com/articles/super-bowl-2026-much-does-170000917.html>

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